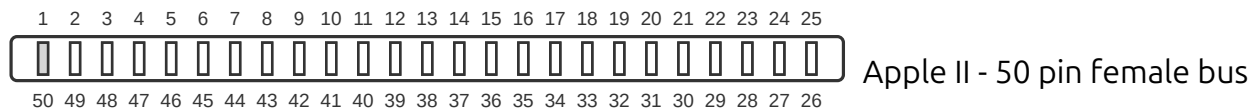


Apple II Slot

50-pin double-sided edge connector expansion slot used in Apple II, II+, IIe, and IIGS computers, carrying the 6502 address, data, and clock buses.



connector

The Apple II peripheral expansion slot is a proprietary 50-pin double-sided edge connector (25 contacts per side, 100 mil pitch, 200 mil row spacing) introduced with the Apple II in 1977 and retained, with minor signal changes, through the Apple IIe and Apple IIGS. It exposes the 6502-family bus directly: a 16-bit address bus (A0–A15), an 8-bit bidirectional data bus (D0–D7), phase clocks, interrupt and DMA daisy-chain lines, and four power rails (+5 V, −5 V, +12 V, −12 V). Up to seven slots are provided in the IIe (Slots 1–7); the IIGS retains the same seven slots with full backward compatibility. Several pin assignments differ between models — most notably pins 19, 35, and 39 — so card designers must account for the target platform.

Pin	Name	Direction	Description
1	/IOSEL	OUT	I/O Select. Active when page \$Cn gets accessed. N.C. on slot 0
2	A0	IN/OUT	Buffered address bus
3	A1	IN/OUT	Buffered address bus
4	A2	IN/OUT	Buffered address bus
5	A3	IN/OUT	Buffered address bus
6	A4	IN/OUT	Buffered address bus
7	A5	IN/OUT	Buffered address bus
8	A6	IN/OUT	Buffered address bus
9	A7	IN/OUT	Buffered address bus

Pin	Name	Direction	Description
10	A8	IN/OUT	Buffered address bus
11	A9	IN/OUT	Buffered address bus
12	A10	IN/OUT	Buffered address bus
13	A11	IN/OUT	Buffered address bus
14	A12	IN/OUT	Buffered address bus
15	A13	IN/OUT	Buffered address bus
16	A14	IN/OUT	Buffered address bus
17	A15	IN/OUT	Buffered address bus
18	R/W	IN/OUT	Buffered Read/Write signal.
19	SYNC	OUT	Only Slot 7. SYNC from Video Generator. Not on Rev 0 Boards. Testpin on Slot 1 for //e
20	/IOSTRB	OUT	I/O Strobe. Active when \$C800 and \$CFFF gets accessed
21	/RDY	IN	Activation during Phi1 will halt the <u>CPU</u> , with the address bus holding the last address
22	/DMA	IN	Activation disables the 6502's address bus and halts the CPU
23	/INTOUT	IN	Daisy-chained interrupt output to lower priority devices
24	/DMAOUT	IN	Daisy-chained DMA output to lower priority devices
25	+5V		+5 Volt power supply. Max. 500mA for ALL peripheral boards
26	<u>GND</u>		System electrical ground
27	/DMAIN	OUT	Daisy-chained DMA input from higher priority devices
28	/INTIN	OUT	Daisy-chained interrupt input from higher priority devices
29	/NMI	IN	Non-Maskable Interrupt. Monitor <u>ROM</u> starts interrupt handling routine at location \$3FB
30	/IRQ	IN	Interrupt ReQuest. Monitor starts the routine pointed to by \$3FE/F

Pin	Name	Direction	Description
31	/RES	IN	RESet
32	/INH	IN	INHibits the on board ROMs (\$D000-\$FFFF)
33	-12V		-12 Volt power supply. Max. 200mA for ALL peripheral boards
34	-5V		-5 Volt power supply. Max. 200mA for ALL peripheral boards
35	COLORREF	OUT	Only Slot 7. 3.5 MHz Video COLOR REF. Not on Rev 0 Boards. Testpin on Slot 1 for //e. M2B0 on A2gs
36	7M	OUT	7Mhz clock
37	Q3	OUT	2Mhz asymetrical clock
38	PHI1	OUT	1 MHz phase 1 clock
39	Various	OUT	USER1 on A2: Disable adressdecode. 65C02 SYNC on A2e. M2SEL on A2gs
40	PHI0	OUT	1 MHz phase 0 clock (Inverted PHI1)
41	/DEVSEL	OUT	DEVElECT. Active when \$C0nX gets accessed; n - Slot#+8
42	D7	IN/OUT	Buffered bi-directional data bus
43	D6	IN/OUT	Buffered bi-directional data bus
44	D5	IN/OUT	Buffered bi-directional data bus
45	D4	IN/OUT	Buffered bi-directional data bus
46	D3	IN/OUT	Buffered bi-directional data bus
47	D2	IN/OUT	Buffered bi-directional data bus
48	D1	IN/OUT	Buffered bi-directional data bus
49	D0	IN/OUT	Buffered bi-directional data bus
50	+12V		+12 Volt power supply. Max. 250mA for ALL peripheral boards

Notes

- Signal names prefixed with / (or shown with a leading slash in the table) are **active-low**.
- The slot is a **25-finger double-sided edge connector**; the component side carries odd-numbered pins and the solder side carries even-numbered pins by convention.
- Power budgets are cumulative across all installed cards: +5 V is limited to 500 mA total, −5 V and −12 V to 200 mA each, and +12 V to 250 mA total for all peripheral boards.
- **Slot-specific signals** (pins 19, 35, 39) vary by model and by slot number:
 - Pin 19 carries SYNC (horizontal/vertical video sync) only on Slot 7 of the original Apple II; on the IIe it becomes a test point on Slot 1.
 - Pin 35 carries the 3.58 MHz colour-reference signal (COLORREF) on Slot 7 of the Apple II/IIe; on the IIGS it is reassigned to M2B0.
 - Pin 39 is USER1 (address-decode disable) on the Apple II, 65C02 SYNC on the IIe, and M2SEL on the IIGS.
- The /INH signal inhibits the on-board ROMs in the \$D000–\$FFFF range, allowing peripheral cards to substitute their own ROM.
- DMA and interrupt priority are resolved by **daisy-chaining**: /DMAIN/DMAOUT and /INTIN/INTOUT pass priority from higher-numbered to lower-numbered slots.
- The Apple IIGS retains the same 50-pin edge connector and maintains backward compatibility with Apple IIe peripheral cards.

Signals

- **/IOSEL** — I/O Select; asserted when page \$Cn is accessed (where n = slot number); not connected on Slot 0.
- **A0–A15** — Buffered 16-bit address bus from the 6502-family CPU.
- **R/W** — Buffered Read/Write signal; high = read, low = write.
- **SYNC** — (Slot 7 only, Apple II/II+) SYNC output from the video generator; used as a test point on Slot 1 of the IIe; absent on Revision 0 boards.
- **/IOSTRB** — I/O Strobe; asserted when addresses \$C800–\$CFFF are accessed.
- **/RDY** — Ready; when asserted during PHI1, halts the CPU with the address bus frozen at the last address.
- **/DMA** — Direct Memory Access request; disables the 6502 address bus and halts the CPU, allowing a peripheral to take bus control.
- **/INTOUT** — Daisy-chained interrupt output passed to lower-priority (higher-numbered) devices.
- **/DMAOUT** — Daisy-chained DMA output passed to lower-priority devices.
- **+5V** — +5 V power rail; maximum 500 mA shared across all peripheral cards.
- **GND** — System ground.
- **/DMAIN** — Daisy-chained DMA input received from higher-priority devices.

- **/INTIN** — Daisy-chained interrupt input received from higher-priority devices.
- **/NMI** — Non-Maskable Interrupt; the Monitor ROM begins interrupt handling at \$3FB.
- **/IRQ** — Interrupt ReQuest; the Monitor begins the routine pointed to by \$3FE/\$3FF.
- **/RES** — Reset signal.
- **/INH** — Inhibit; disables on-board ROMs in the \$D000–\$FFFF range.
- **–12V** — –12 V power rail; maximum 200 mA shared across all peripheral cards.
- **–5V** — –5 V power rail; maximum 200 mA shared across all peripheral cards.
- **COLORREF** — (Slot 7 only, Apple II/Ile) 3.58 MHz colour reference; absent on Revision 0 boards; reassigned to M2B0 on the IIGS.
- **7M** — 7 MHz master clock output.
- **Q3** — 2 MHz asymmetrical clock output.
- **PHI1** — 1 MHz phase-1 clock.
- **Various (pin 39)** — USER1 (address-decode disable) on the Apple II; 65C02 SYNC on the IIe; M2SEL on the IIGS.
- **PHI0** — 1 MHz phase-0 clock (inverted PHI1).
- **/DEVSEL** — Device Select; asserted when \$C0nX is accessed, where n = slot number + 8.
- **D0–D7** — Buffered bidirectional 8-bit data bus.
- **+12V** — +12 V power rail; maximum 250 mA shared across all peripheral cards.